Deep Insights, Smart Moves: Advancing Traffic Prediction with Hybrid LSTM-GCN Architectures

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Introduction

In this study, we have focused on preprocessing the Beijing-sevenpart, PEMS, and T-drive trajectory datasets, which has been a cornerstone in developing our traffic prediction model. The integration of LSTM and GCN has proven to be a robust approach for forecasting traffic patterns, and the optimization of the Dijkstra algorithm with dynamic weight adjustment has significantly improved path planning. This article details our methodologies and findings.

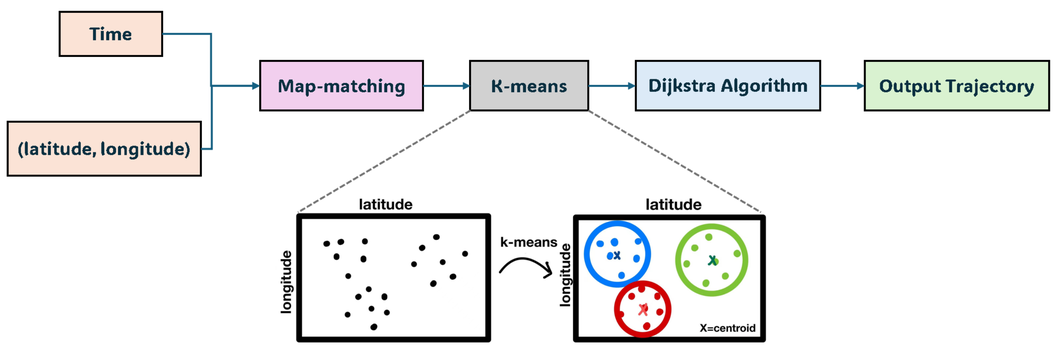
Data processing

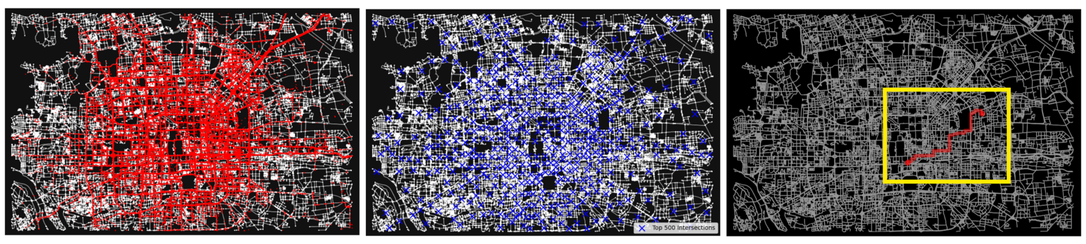
We conducted extensive data preprocessing on multiple datasets, including Microsoft T-Drive, PeMS, and Beijing_sevenpart, to ensure high data quality and consistency. This comprehensive process involved rigorous cleaning, integration, and feature engineering, such as removing anomalies, filling in missing data, and encoding GPS points with road networks using tools like osmnx. Additionally, we integrated various data sources and applied clustering techniques to analyze traffic patterns in relation to their geographical context, thus providing a robust foundation for our predictive models.

Model development

1. Optimal Path Planning

- Integrated map-matching to accurately align GPS trajectories onto the road network, using the osmnx library.
- Applied K-means clustering with an initial random selection of 500 centroids to classify landmarks.
- Optimized path calculation using Dijkstra's algorithm, where edge weights represented the Euclidean distance between nodes.





2. Traffic Trajectories Prediction Based on LSTM

Operations on Dataloader

(1) To train the LSTM model, we convert the raw data into sequential data. Each sequence contains 5 consecutive coordinate points used to predict the next coordinate point.

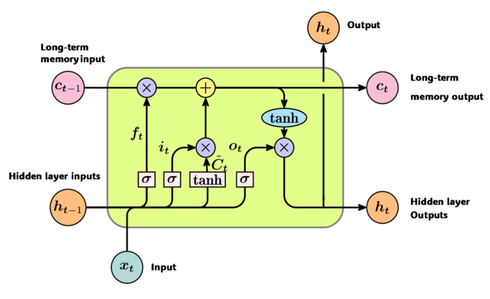
(2) Due to the scale sensitivity of LSTM to input data, we use MinMaxScaler to standardize the data within the range of 0 to 1.

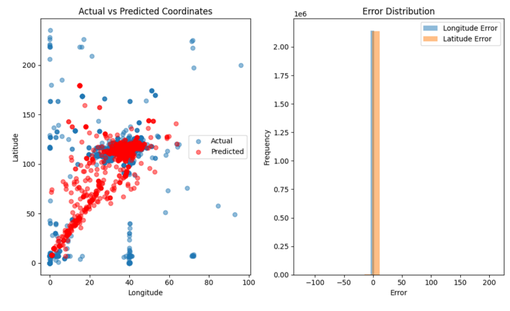
Hyperparameter fine-tuning

During the model training process, we attempted different hyperparameter configurations, including the number of units in the LSTM layer and the number of epochs trained. We ultimately selected **50 units and 5 epochs**, and in the test set, we also verified the rationality of the hyperparameter configuration

Model evaluation

To ensure the generalization of the model, we need to ensure that the test set consists of data that did not participate in training. Therefore, we divided the training data into an **80 percent training set** and a **20 percent testing set** to monitor the change pattern of the loss function. We evaluated the metrics of MSE and RMSE, and observed a continuous decreasing trend in the loss function during the experiment.





Evaluation Metric	LSTM
MSE	1.98724
RMSE	1.40970

3. Traffic Flow Prediction Based on GCN

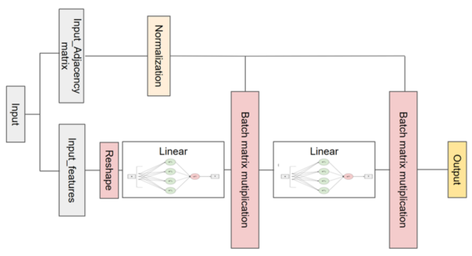
Below is the structure of GCN model. GCN (Graph Convolutional Networks) is an advanced neural network architecture specifically designed to process graph-structured data. Its core advantage is the ability to update the representation of its own nodes by aggregating information from neighboring nodes, a process that enables neural networks to capture and understand complex dependencies between nodes. The update rule for GCNs can be expressed as:

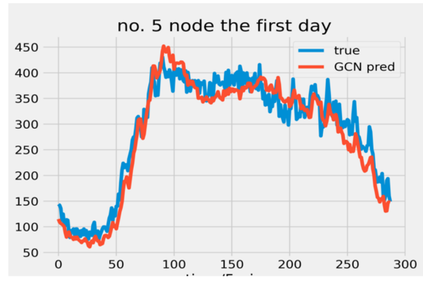
$$h_i^{(l+1)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \frac{1}{c_{ij}} W^{(l)} h_j^{(l)}\right)$$

where $h_i^{(l)}$ represents the feature representation of node i at layer l, $\mathcal{N}(i)$ is the set of neighboring nodes of node i, c_{ij} is the normalization coefficient, $W^{(l)}$ is the weight matrix at layer l, and σ is the activation function.

Hyperparameter configuration We use the Adam optimizer for training with a learning rate of 0.03. Train the model for 30 epoches. To ensure the reproducibility of the experiment, we used the same random seeds in all experiments.

Model evaluation We evaluated the MSE, RMSE, MAPE, and loss values on the testing machine. During the experiment, we found that all four showed a significant decrease, indicating that our model is reasonable for predicting traffic flow. We have presented the results of 30 epochs in Table.





Evaluation Metric	GCN
Testloss	0.0081
MSE	0.0645
RMSE	0.0900
MAPE	0.1279%

4. Dynamic Time Weight Adjustment

- Enhanced Dijkstra's algorithm by introducing a dynamic weight adjustment mechanism based on real-time traffic speed data.
- Calculated initial weights using the formula $w_{ij} = \frac{d_{ij}}{v_{ij}}$, where d_{ij} is the distance and v_{ij} is the average speed.
- Adjusted weights at every time step based on updated speed data, allowing the algorithm to adapt to traffic conditions with a time complexity of $O(|V|^2)$.

Practical Applications

Our traffic prediction model offers extensive practical benefits, enhancing urban traffic management by optimizing traffic signal control and providing real-time path planning for autonomous vehicles. It improves travel efficiency, safety, and the environment by reducing emissions. Analyzing traffic data, the model identifies congestion points and suggests optimal routes, advancing transportation technology.

Conclusions

This research harnesses the power of deep learning technologies to tackle intricate urban traffic challenges, showcasing their potential to drive the evolution of intelligent transportation systems and reshape urban planning and management. By meticulously preprocessing the Beijing_sevenpart, PEMS, and T-drive trajectory datasets, we established a solid data foundation through cleaning, integration, and feature engineering. Building on this, we developed an advanced traffic prediction model that integrates LSTM for its prowess in time series analysis to capture traffic flow dynamics and long-term dependencies, complemented by GCN to boost prediction accuracy by examining the road network's topological structure. Additionally, we refined the Dijkstra algorithm for dynamic weight adjustment, enabling our model to respond to real-time traffic conditions and optimize path planning. Our model, validated across various datasets, has proven to be highly accurate and dependable in traffic flow prediction and path planning. This innovation not only pushes technological boundaries but also offers substantial practical potential and value, significantly enhancing the intelligence of urban transportation systems and bolstering urban traffic management through its predictive and planning capabilities.